Project Report

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Semester: 4th
Date of Performance: 19/04/2026
Subject Name: Internet of Things Lab
Subject Code: 24CAP-751

Aim: IoT Based Smart WiFi Analyzer System to detect, monitor, and analyze nearby wireless networks in real-time.

Objectives:

1. To scan and analyze nearby WiFi networks using ESP32.
2. To measure signal strength (RSSI) and network quality.
3. To identify secure and open networks.
4. To display real-time data on web dashboard and mobile application (Blynk).

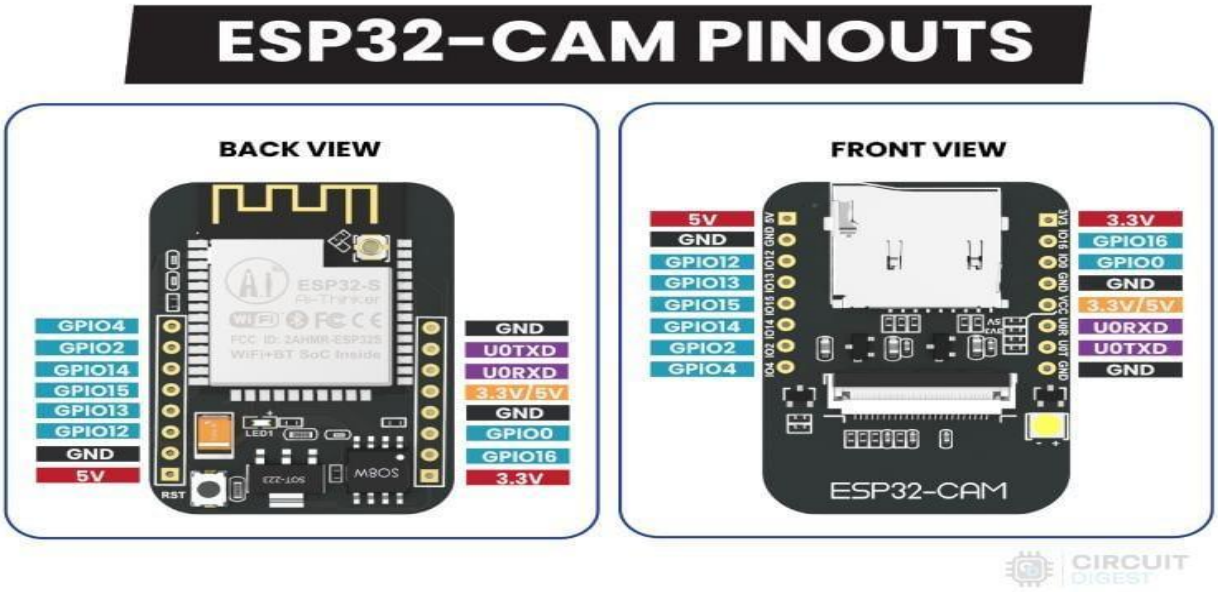
Components Required:

Sno	Name of Component	Qty.
1.	ESP32-CAM Module	1
2.	FTDI Programmer	1
3.	Jumper Wires	As required
4.	Breadboard	1

Details of Components:

1. ESP32-CAM Module

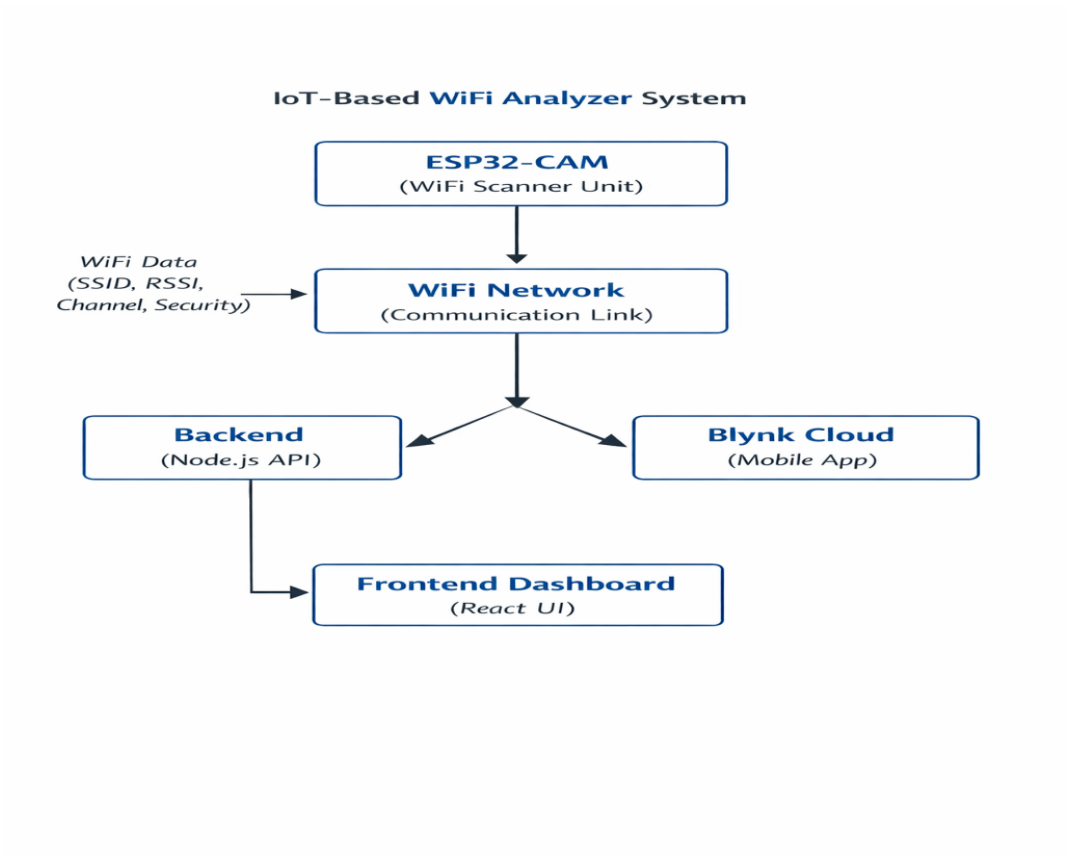
The ESP32-CAM is a low-cost microcontroller with built-in WiFi capability, widely used in IoT applications. It is based on the ESP32 chip, which supports wireless communication and data processing. In this project, the ESP32-CAM is used to scan nearby WiFi networks and collect important parameters such as SSID, signal strength (RSSI), channel, and encryption type. Although it includes a camera module, the camera feature is not used in this system. The ESP32 acts as the main processing unit and sends data to the backend server and mobile dashboard.



2. FTDI Programmer (USB to Serial Converter)

The FTDI programmer is used to upload code from the computer to the ESP32-CAM module. Since ESP32-CAM does not have a built-in USB port, the FTDI module provides serial communication between the computer and the ESP32. It converts USB signals into serial signals, enabling programming and debugging through the Arduino IDE.

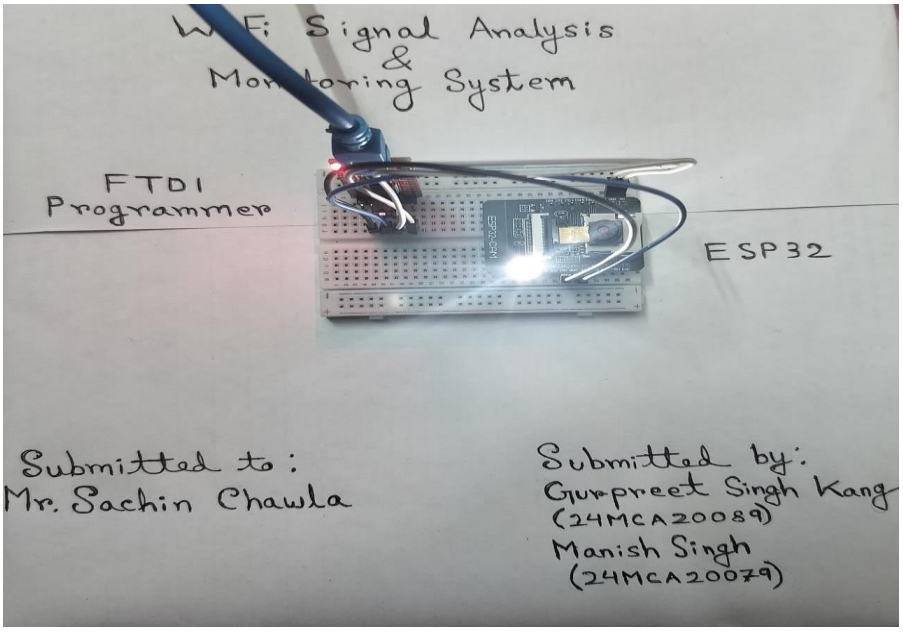
Block Diagram of Designed Model:



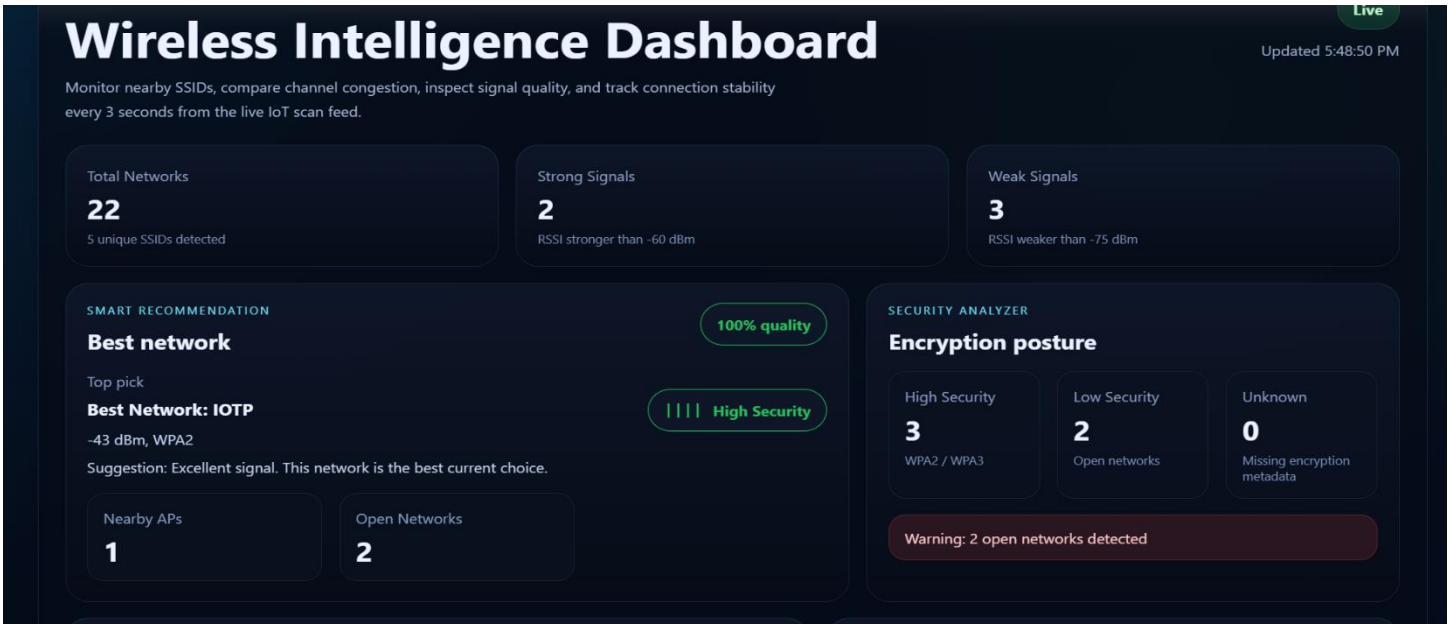
Working of Designed Model:

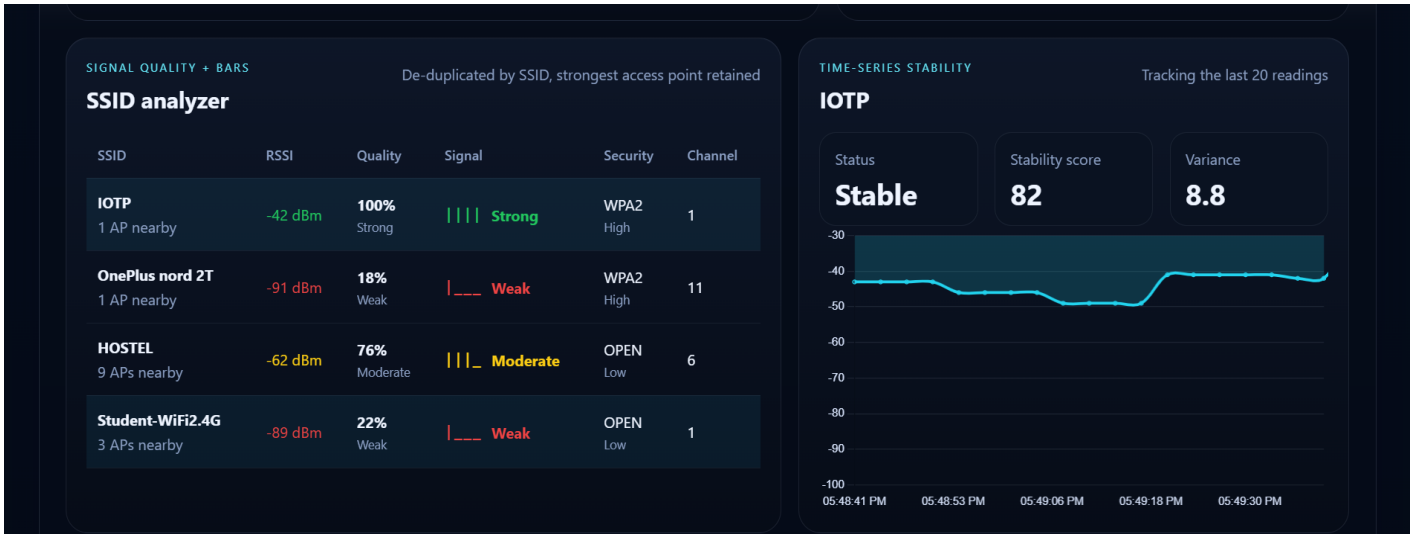
- The **ESP32-CAM** acts as the main sensing and processing unit. It scans nearby WiFi networks and collects data such as SSID, signal strength (RSSI), channel, and encryption type.
- The collected data is transmitted over the **WiFi network**, which acts as a communication medium between the ESP32 and other system components.
- The data is sent to a **backend server (Node.js)** using HTTP requests. The backend processes and stores the data and makes it available to the frontend.
- The **frontend dashboard (React)** displays the analyzed data in graphical form, including signal strength, best network, and channel interference.
- At the same time, important summarized data is sent to the **Blynk Cloud**, which allows monitoring through a mobile application.

Pictures of Prototype:

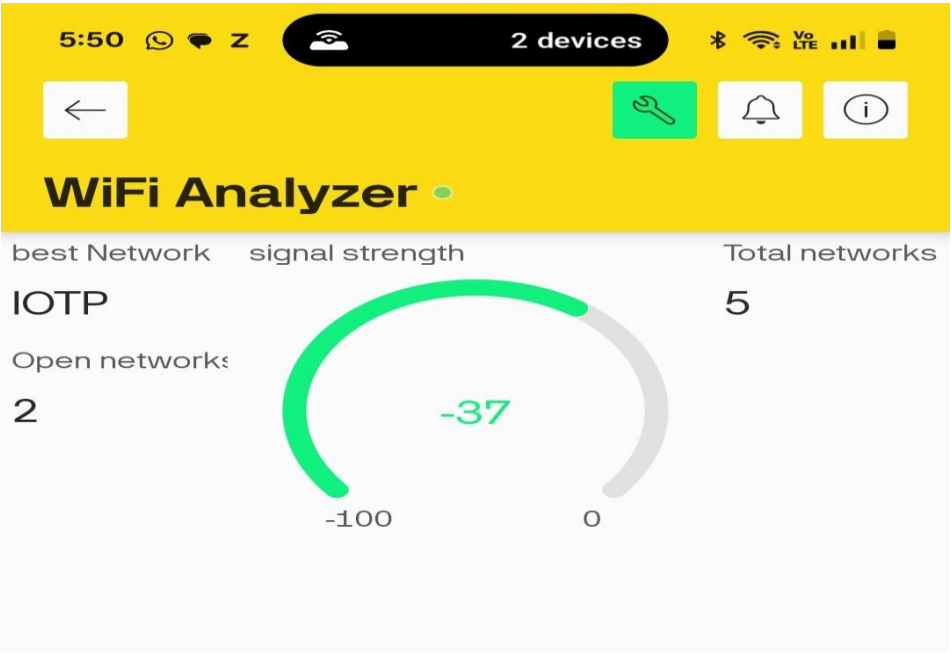


Output of Deigned Model/Prototype





BLYNK APP (OUTPUT)



Learning outcomes :

1. Learned how to use ESP32 for IoT applications.
2. Gained knowledge of WiFi networking and signal analysis.
3. Developed backend using Node.js and frontend using React.
4. Integrated IoT system with mobile app (Blynk).